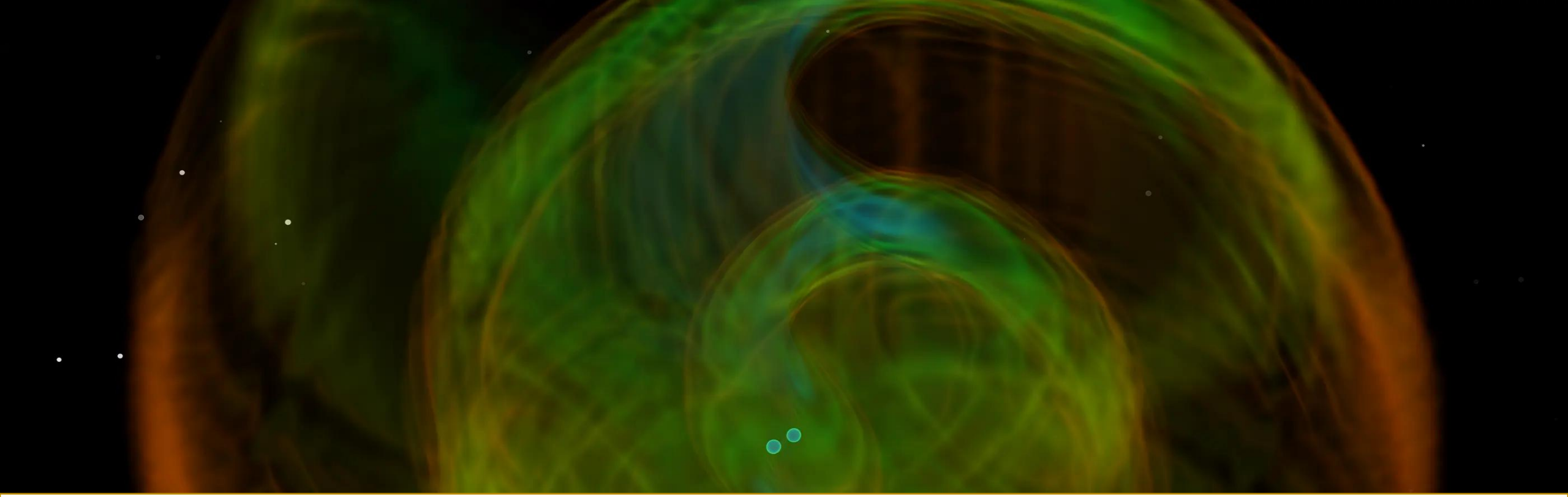


SEARCH FOR THE GW190425 OPTICAL COUNTERPART IN THE ZTF SURVEY

Efremova M. A., Pozanenko A. S., Pankov N. S., Volnova A. A.

YSC 2026

GW190425 VS GW170817

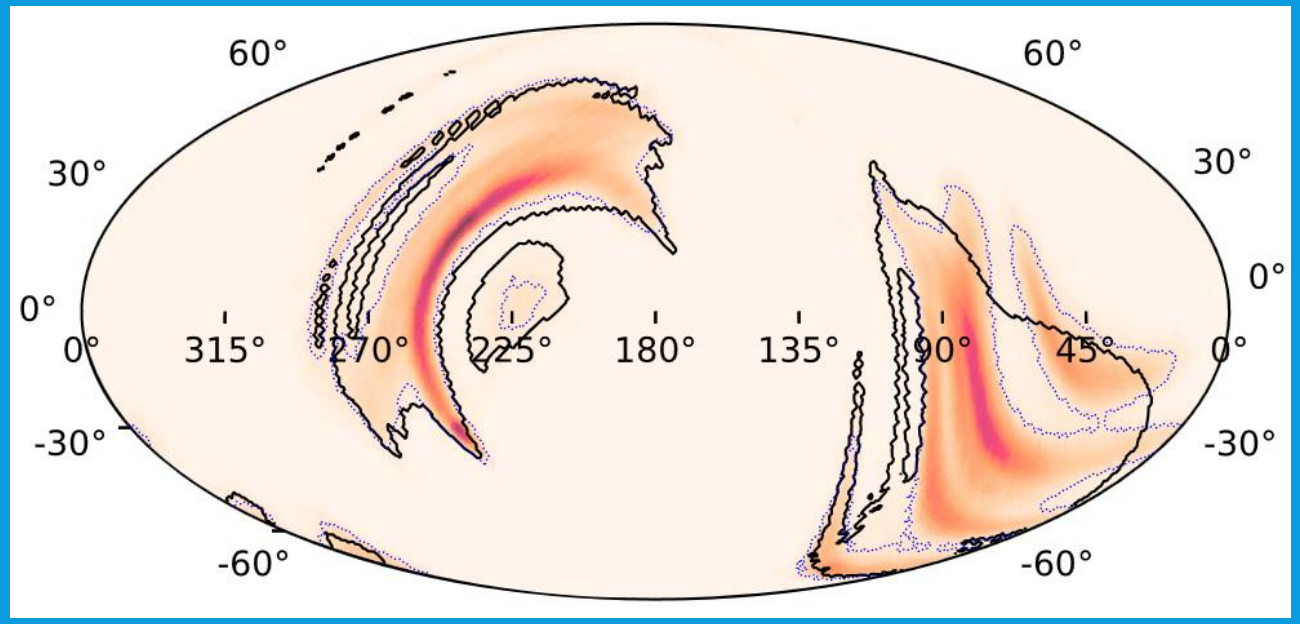
Name:	GW190425	GW170817
Distance:	159^{+69}_{-71} Mpc	$43.8^{+2.9}_{-6.9}$ Mpc
Total Mass:	$3.4^{+0.3}_{-0.1}\ M_{\odot}$	$2.74^{+0.04}_{-0.01}\ M_{\odot}$
Apparent Magnitude:	NOT DETECTED	17.46 ± 0.03 (r-band)

[S. Valenti et al., 2017]

[B. P. Abbott et al., 2020] [B. P. Abbott et al., 2017]

LOCALIZATION AREAS

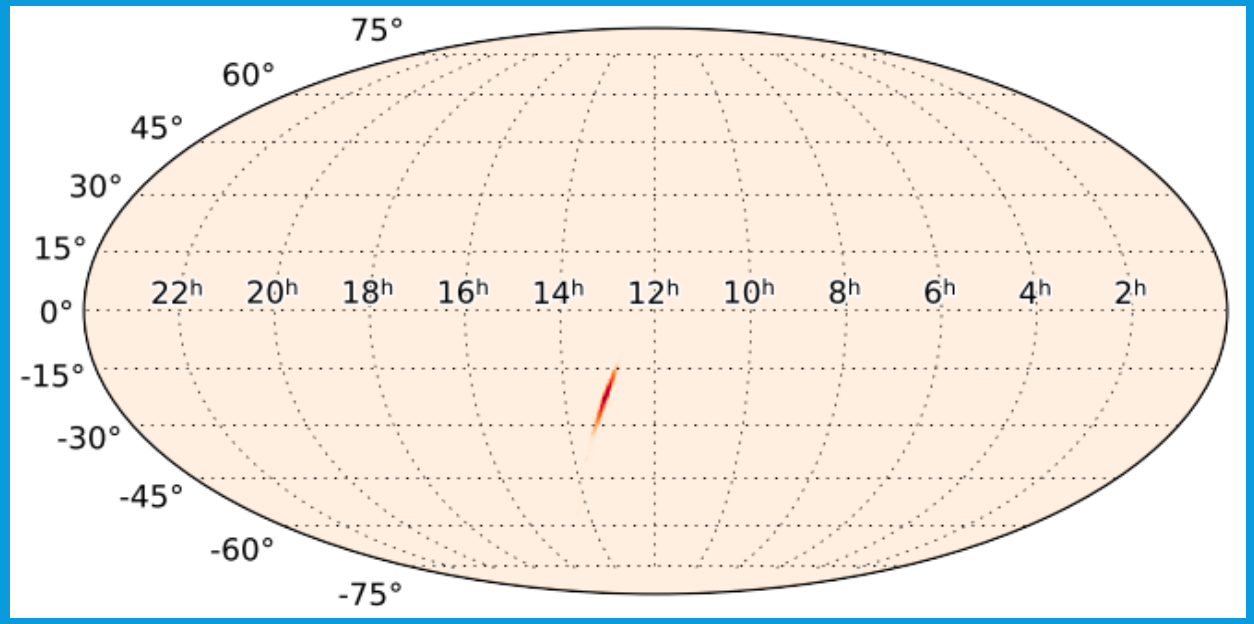
GW190425



8284 deg^2

[B. P. Abbott et al., 2020]

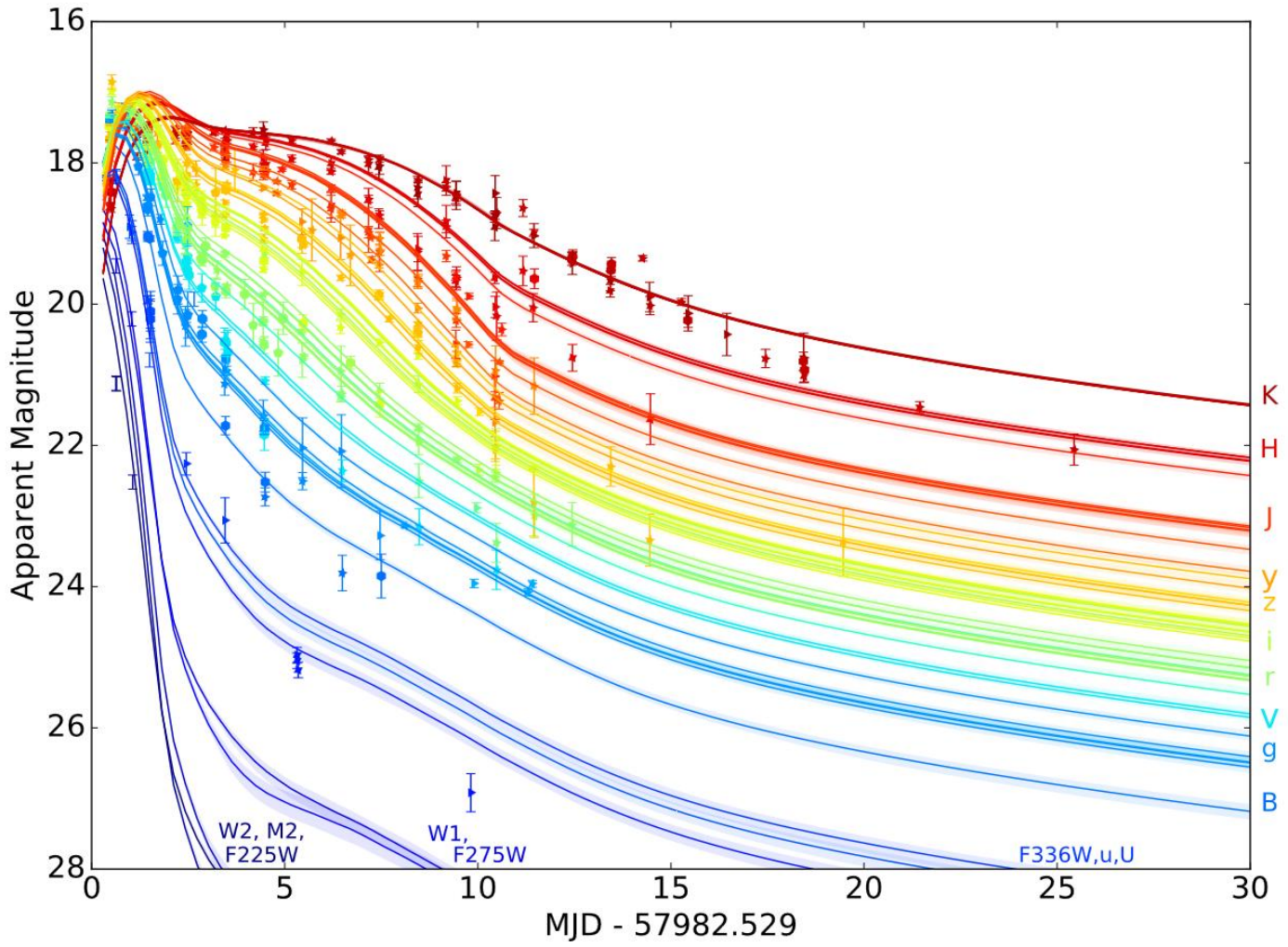
GW170817



28 deg^2

[B. P. Abbott et al., 2017]

GW170817 LIGHT CURVE



- GW190425 would be of 20.3 mag

(assuming all parameters but distance to be the same as of GW170817)

[Villar et al., 2017]

ZTF ALERT STREAM

© Palomar/Caltech

ZTF facts:

- Cadence of ~2 days
- 5σ detection limit of 20.7 mag
- Raw data is processed in real time
- Alerts are freely available immediately

- A ZTF transient alert is defined as a 5σ change in brightness in the image relative to the reference epoch

PREVIOUS WORK

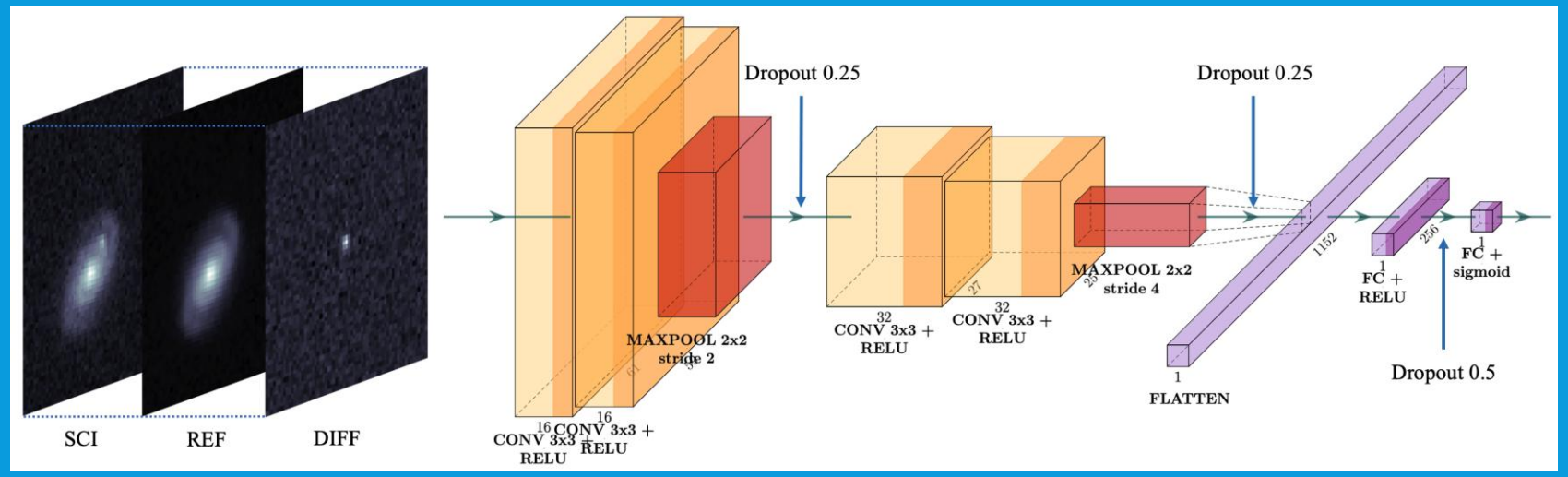
GROWTH on S190425z: Searching thousands of square degrees to identify an optical or infrared counterpart to a binary neutron star merger with the Zwicky Transient Facility and Palomar Gattini IR

Michael W. Coughlin,[1] Tomás Ahumada,[2] Shreya Anand,[1] Kishalay De,[1] Matthew J. Hankins,[1]

- 338 646 ZTF transient "alerts" over the first two nights after trigger
- None of the alerts turned out to be a viable counterpart

OUR METHOD

- Check all difference data and not only alerts
- Create a combined galaxy catalog
- Use galaxy cutouts
- The threshold is lower than in the previous work ($5\sigma \rightarrow 3\sigma$)
- Apply ML algorithm to score real/bogus for objects on difference images [Duev+ 2019]



GALAXY CATALOG

- Combined from SDSS-DR16, DESI-DR8, GLADE+, NED-LVS, CatWISE and GAIA-DR3

- Filtered to include only objects:
  - in the localization area
  - similar to galaxies
  - with a distance either in the range of GW190425, or unknown

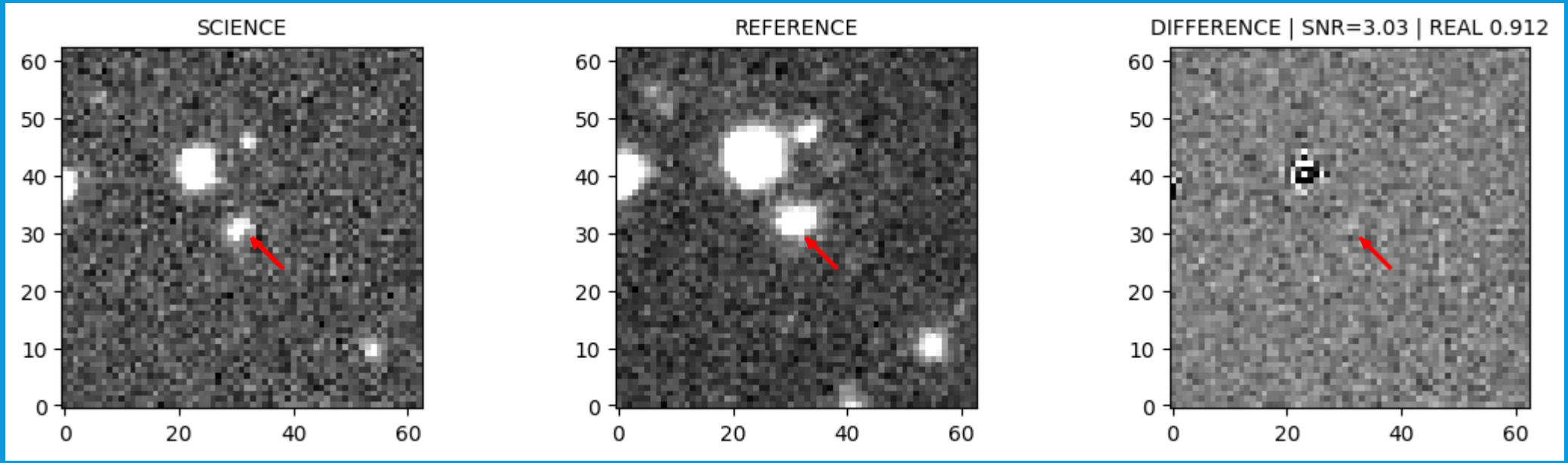
Total number of galaxies:	37 118
Galaxies with known distances:	273

ZTF IMAGES

- 25-26 of April: ~78k difference image cutouts
- >500 cutouts with SNR ≥ 3 at the possible host galaxy
- Issue: ~4k masked images

CURRENT RESULTS AND PLANS

- Finish analysis of all cutouts
- Obtain a list of candidates



So far, the only candidate: z_phot=2.64±0.89 (i.e. within 159 Mpc); not seen at pre-trigger epochs

QUESTIONS

CANDIDATE EXAMINATION

if SNR ≥ 3:

- pass to ML algorithm for a real/bogus score;

  if REAL:

  - double check through various catalogs;

    if the object is a galaxy and z is in the correct range / unknown:

    - compare its FWHM to star FWHM;

      if smaller, reject as cosmic ray, else:

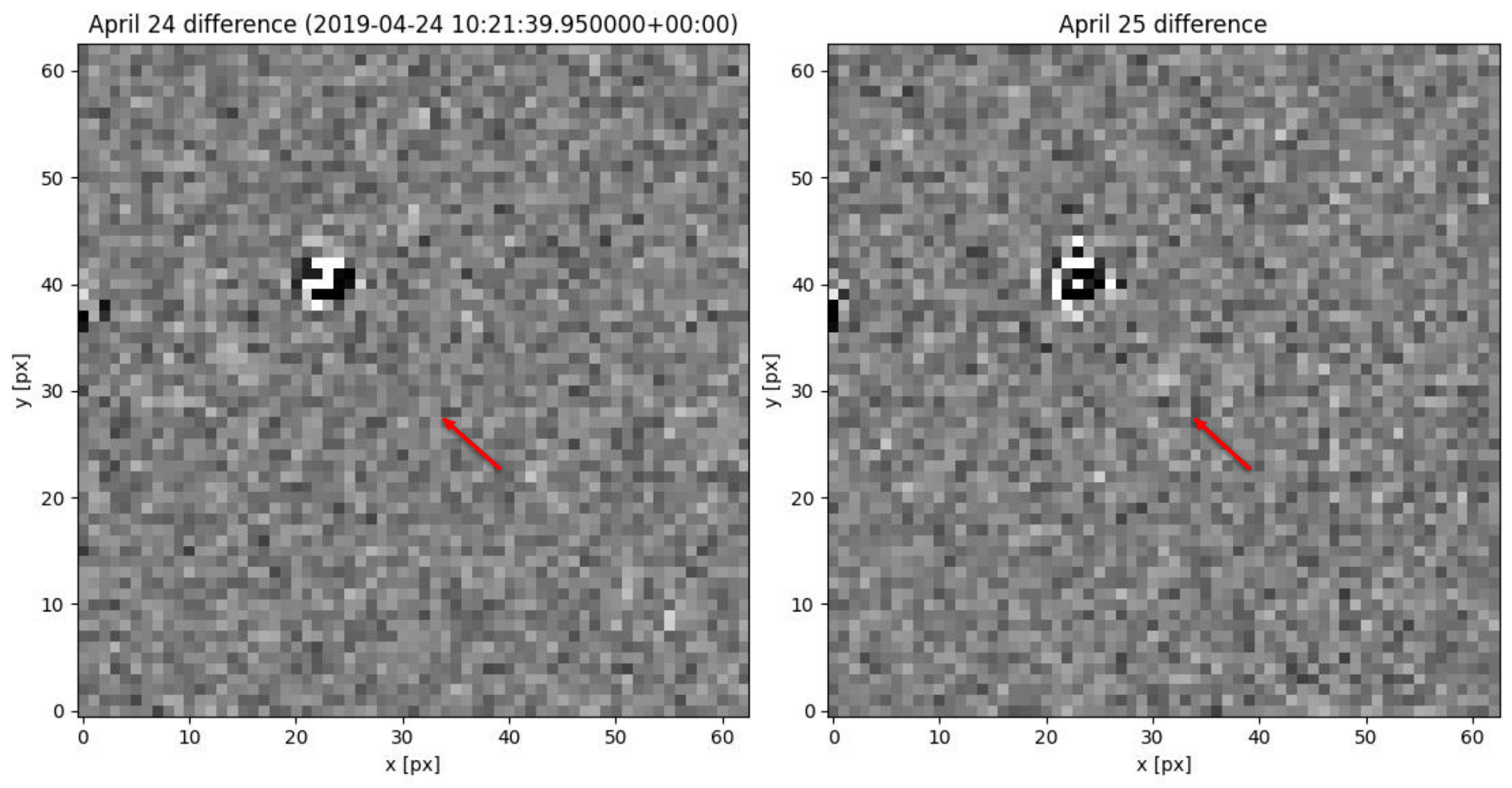
      - check if the object is present pre-trigger;

        if not, check further post-trigger images and study the brightness

WHY ARE SOME CATALOG ENTRIES STARS?

grid study wink redo north hdr multiview match Search

_RAJ2000...	_DEJ2000...	_V	objID	RA_ICRS...	DE_ICRS...	m	c	c	e_RA_IC...
247.69693423	12.93611359	VizieR	12376655311897893...	247.696934	12.936114	3	6	0	0.003
247.69694491	12.9361265	VizieR	12376655311897893...	247.696945	12.936127	3	6	0	0.003
247.6969449	12.93612656	VizieR	12376655311897893...	247.696945	12.936127	4	6	0	0.003
247.69693154	12.9361142	VizieR	12376655311898545...	247.696932	12.936114	3	6	0	0.003
247.69694262	12.93612768	VizieR	12376655311898545...	247.696943	12.936128	3	6	0	0.003
247.69694468	12.93615297	VizieR	12376655311898545...	247.696943	12.936128	2	6	0	0.003
247.69693081	12.93613552	VizieR	12376655671582399...	247.696931	12.936136	3	6	0	0.003
247.69692753	12.93613603	VizieR	12376655671582399...	247.696928	12.936136	3	6	0	0.003
247.69692975	12.93616097	VizieR	12376655671582399...	247.696928	12.936136	1	6	1	0.003
247.6969255	12.93613965	VizieR	12376655671583051...	247.696926	12.93614	3	0	0	0.003
247.69698914	12.93611552	VizieR	12376655671583051...	247.696989	12.936116	4	0	0	0.002

- Multiple SDSS entries per one object
- Some of them can pass the filtering

DIFFERENCE IMAGES FOR CANDIDATE